

UNIT – 5
ASSESSMENT – 1
REPORT

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1. Aim

To design a hybrid AI recommendation engine using PySpark RDDs by combining content-based and collaborative filtering techniques.

Objectives

1. Process large-scale user-item interaction data.
2. Implement content-based filtering.
3. Implement collaborative filtering.
4. Combine both approaches into a hybrid recommendation.
5. Use RDD transformations for distributed processing.
6. Adapt recommendations using an intelligent agent.

Algorithm

1. Create the Spark context.
2. Load user-item interaction data into an RDD.
3. Load item-category information.
4. Select the target user.
5. Identify items liked by the target user.
6. Perform content-based filtering.

7. Identify users with similar preferences.
8. Perform collaborative filtering.
9. Combine both recommendation lists.
10. Receive user feedback.
11. Allow the intelligent agent to modify recommendation priorities.

Result

The hybrid recommendation engine successfully combines content-based and collaborative filtering using PySpark RDDs and produces personalized recommendations.

2. Aim

To develop a distributed disaster detection system using PySpark RDDs and intelligent agents.

Objectives

- Process multi-source disaster data.
- Apply RDD transformations.
- Fuse sensor, satellite and social media information.
- Detect disaster events.
- Generate early warnings.
- Use intelligent agents for response decisions.

Algorithm

1. Initialize Spark.
2. Load sensor data.
3. Load satellite observations.

4. Load social media reports.
5. Filter high-intensity observations.
6. Count social media reports.
7. Combine evidence using RDD operations.
8. Calculate disaster confidence.
9. Generate alerts when evidence exceeds the threshold.
10. Allow the intelligent agent to recommend a response.

Result

The distributed system successfully combines information from multiple sources and identifies potential disaster events.

3. Aim

To develop a scalable fraud detection framework using PySpark RDDs and intelligent agents.

Objectives

1. Process large-scale transaction data.
2. Detect anomalous transactions.
3. Group suspicious transactions by user.
4. Generate fraud alerts.
5. Use distributed RDD processing.
6. Allow intelligent agents to continuously monitor suspicious activity.

Algorithm

1. Initialize Spark.
2. Load transaction records.
3. Convert records into an RDD.

4. Apply anomaly detection rules.
5. Filter suspicious transactions.
6. Group suspicious transactions by user.
7. Count suspicious transactions.
8. Identify high-risk users.
9. Generate alerts.
10. Allow the fraud detection agent to initiate further investigation.

Result

The system successfully identifies abnormal financial transactions and detects users with repeated suspicious activity.

4. Aim

To construct a distributed AI-based supply chain optimization system using PySpark RDDs and intelligent agents.

Objectives

- Process inventory and demand data.
- Detect inventory shortages.
- Analyze expected demand.
- Generate restocking recommendations.
- Demonstrate distributed RDD processing.
- Use intelligent agents for autonomous decisions.

Algorithm

1. Initialize Spark.
2. Load inventory data.

3. Load demand data.
4. Convert data into RDDs.
5. Combine inventory and demand information.
6. Calculate inventory shortages.
7. Identify products requiring restocking.
8. Use the inventory agent to generate decisions.
9. Recommend required quantities.
10. Update decisions as demand changes.

Result

The system identifies products with insufficient inventory and automatically generates restocking recommendations.

5. Aim

To develop a personalized learning analytics platform using PySpark RDDs and intelligent agents.

Objectives

1. Analyze student performance data.
2. Calculate student-level performance.
3. Identify weak subjects.
4. Generate personalized learning recommendations.
5. Process student data using distributed RDDs.
6. Adapt learning recommendations based on feedback.

Algorithm

1. Initialize Spark.

2. Load student performance data.
3. Create an RDD.
4. Calculate average scores.
5. Identify subjects with low scores.
6. Classify weak learning areas.
7. Generate personalized content recommendations.
8. Receive student feedback.
9. Allow the learning agent to modify recommendations.
10. Repeat the process as new learning data arrives.

Result

The system successfully analyzes student performance and identifies weak subjects. The intelligent learning agent generates personalized learning recommendations according to individual student needs.